TSA6213 SYSTEM ADMINISTRATION & MAINTENANCE

LAB 5

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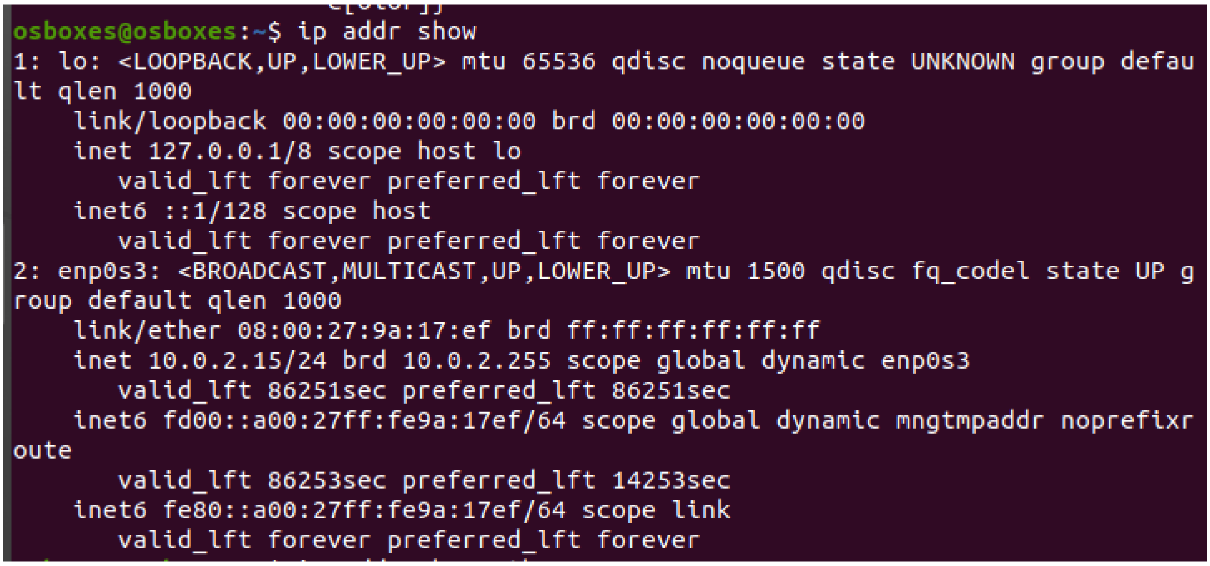
Internet Ready Environment – Network Configuration and Using FTP Server

Network Configuration

1. Understanding how network configuration works under the hood in Linux distributions is invaluable and can come in handy in several scenarios. Typically, Ethernet devices register themselves with names such as ethX, emX, ensX, pXpX, or enpXsY, where X and Y are the device or index numbers, respectively.

a) Use IP command. Find your Ethernet devices, IP address and IPv6 address.

```
> ip
> ip addr show // ip a
```



b) Use hostname command to find IP address

```
> hostname
> hostname -I
> hostnamectl
```

```
osboxes@osboxes:~$ hostname
osboxes
osboxes@osboxes:~$ hostname -I
10.0.2.15 fd00::a00:27ff:fe9a:17ef
osboxes@osboxes:~$ hostnamectl
   Static hostname: osboxes
         Icon name: computer-vm
           Chassis: vm
        Machine ID: 23ee0cd006dd40a3845e22dd5df70f27
           Boot ID: 8ecb32eec630411d806c7e4aeacb50bc
    Virtualization: oracle
  Operating System: Ubuntu 20.04.4 LTS
            Kernel: Linux 5.4.0-200-generic
      Architecture: x86-64
osboxes@osboxes:~$
```

c) Install and use ifconfig command. Find your Ethernet devices, IP address and IPv6 address

> IF command ifconfig not found, intall first

> sudo apt install net-tools

```
osboxes@osboxes:~$ ifconfig

Command 'ifconfig' not found, but can be installed with:

sudo apt install net-tools

osboxes@osboxes:~$ sudo apt install net-tools
[sudo] password for osboxes:
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following NEW packages will be installed:
  net-tools
0 upgraded, 1 newly installed, 0 to remove and 138 not upgraded.
1 not fully installed or removed.
```

> Once installed, run command

> ifconfig

```
osboxes@osboxes:~$ ifconfig
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.0.2.15  netmask 255.255.255.0  broadcast 10.0.2.255
        inet6 fd00::a00:27ff:fe9a:17ef  prefixlen 64  scopeid 0x0<global>
        inet6 fe80::a00:27ff:fe9a:17ef  prefixlen 64  scopeid 0x20<link>
        ether 08:00:27:9a:17:ef  txqueuelen 1000  (Ethernet)
        RX packets 287  bytes 271378 (271.3 KB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 290  bytes 39549 (39.5 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 161  bytes 13989 (13.9 KB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 161  bytes 13989 (13.9 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```

FTP Anonymous Access

2. File Transfer Protocol (FTP) is one of the oldest and most commonly used protocols found on the Internet today. Access to an FTP server can be managed in two ways, either Anonymous or Authenticated. Find and login to any site on the Internet that provide Anonymous FTP login. Use Ubuntu or Debian mirror to find anonymous FTP site.

a) Use text-based login in Linux to access the anonymous FTP site.

> ftp ftp.ubuntu.com

> ftp ftp.us.debian.org

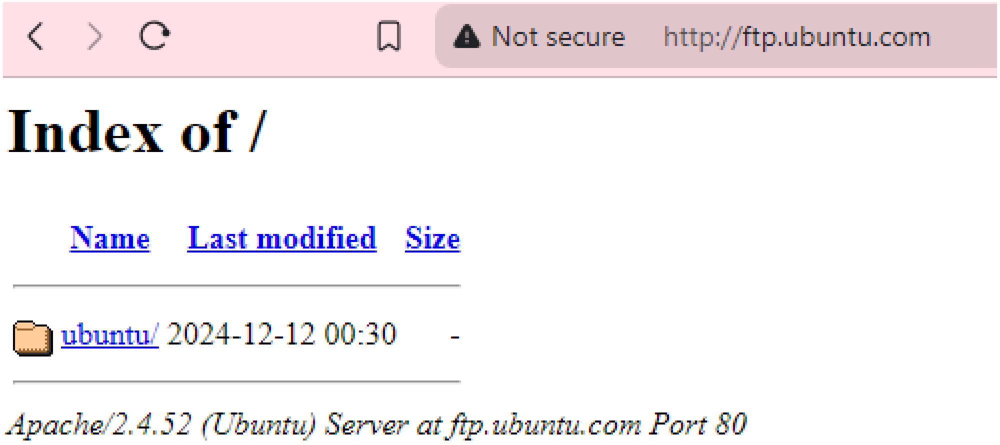
> username: anonymous password: [blank] or email address

> Enter

```
osboxes@osboxes:~$ ftp ftp.ubuntu.com
Connected to ftp.ubuntu.com.
220 FTP server (vsftpd)
Name (ftp.ubuntu.com:osboxes): anonymous
331 Please specify the password.
Password:
230 Login successful.
Remote system type is UNIX.
Using binary mode to transfer files.
ftp>
```

b) Use browser in Windows to access the anonymous FTP site.

> go to `ftp.ubuntu.com` on browser



3. Authenticated Access using ProFtpd FTP server in Windows and Linux

a) Install proftpd as FTP server in your system. Use sudo apt-get install proftpd-basic and use ftp client in terminal to test the installation.

> `sudo apt-get install proftpd-basic`

```
osboxes@osboxes:~$ sudo apt-get install proftpd-basic
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following additional packages will be installed:
  libhiredis0.14 libmemcached11 libmemcachedutil2 proftpd-doc
Suggested packages:
  openbsd-inetd | inet-superserver proftpd-mod-ldap proftpd-mod-mysql
  proftpd-mod-odbc proftpd-mod-pgsql proftpd-mod-sqlite proftpd-mod-geoip
  proftpd-mod-snmp
The following NEW packages will be installed:
  libhiredis0.14 libmemcached11 libmemcachedutil2 proftpd-basic proftpd-doc
0 upgraded, 5 newly installed, 0 to remove and 138 not upgraded.
1 not fully installed or removed.
Need to get 3,509 kB of archives.
After this operation, 9,696 kB of additional disk space will be used.
Do you want to continue? [Y/n] y
```

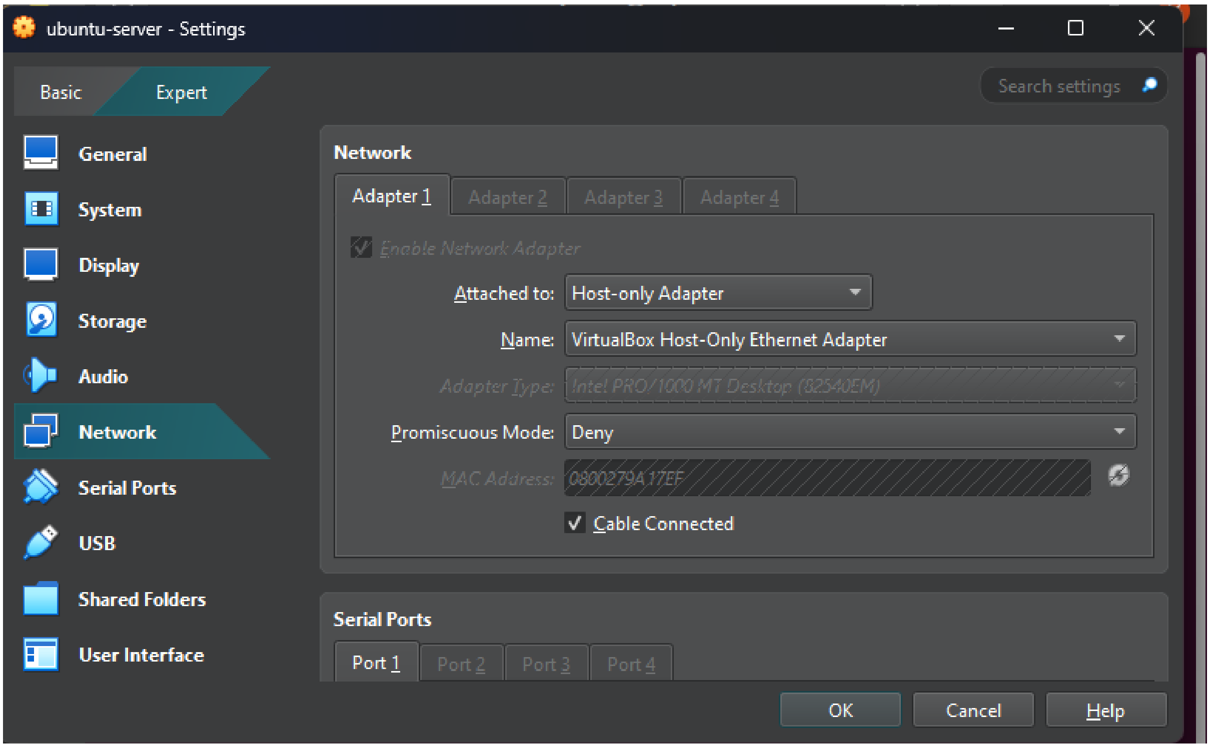
> `ftp ftp.ubuntu.com`

> username: `anonymous` password: [blank]

```
osboxes@osboxes:~$ ftp localhost
Connected to localhost.
220 ProFTPD Server (Debian) [::ffff:127.0.0.1]
Name (localhost:osboxes): osboxes
331 Password required for osboxes
Password:
230 User osboxes logged in
Remote system type is UNIX.
Using binary mode to transfer files.
ftp>
```

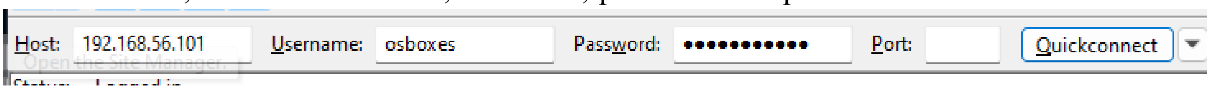
b) Access FTP server from Windows host using any graphical FTP client such as Windows/File Explorer, FileZila, gFTP or any FTP clients. Upload any files from the windows machine to the FTP server.

Go to Machine setting, Network, change the Attached to: NAT to Host-only Adapter

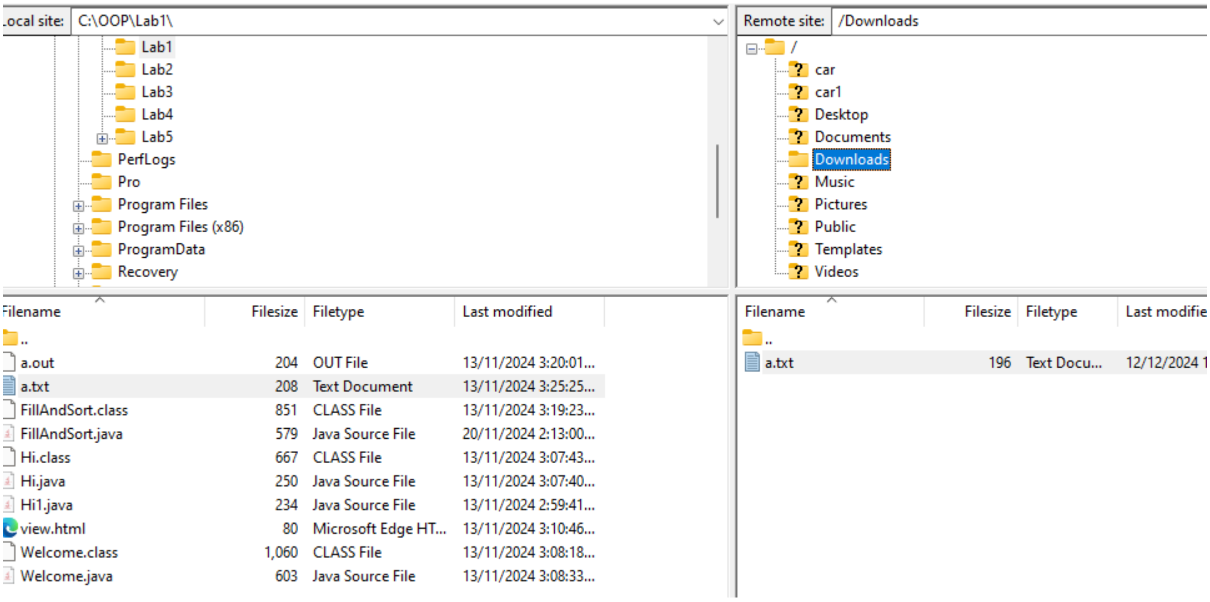


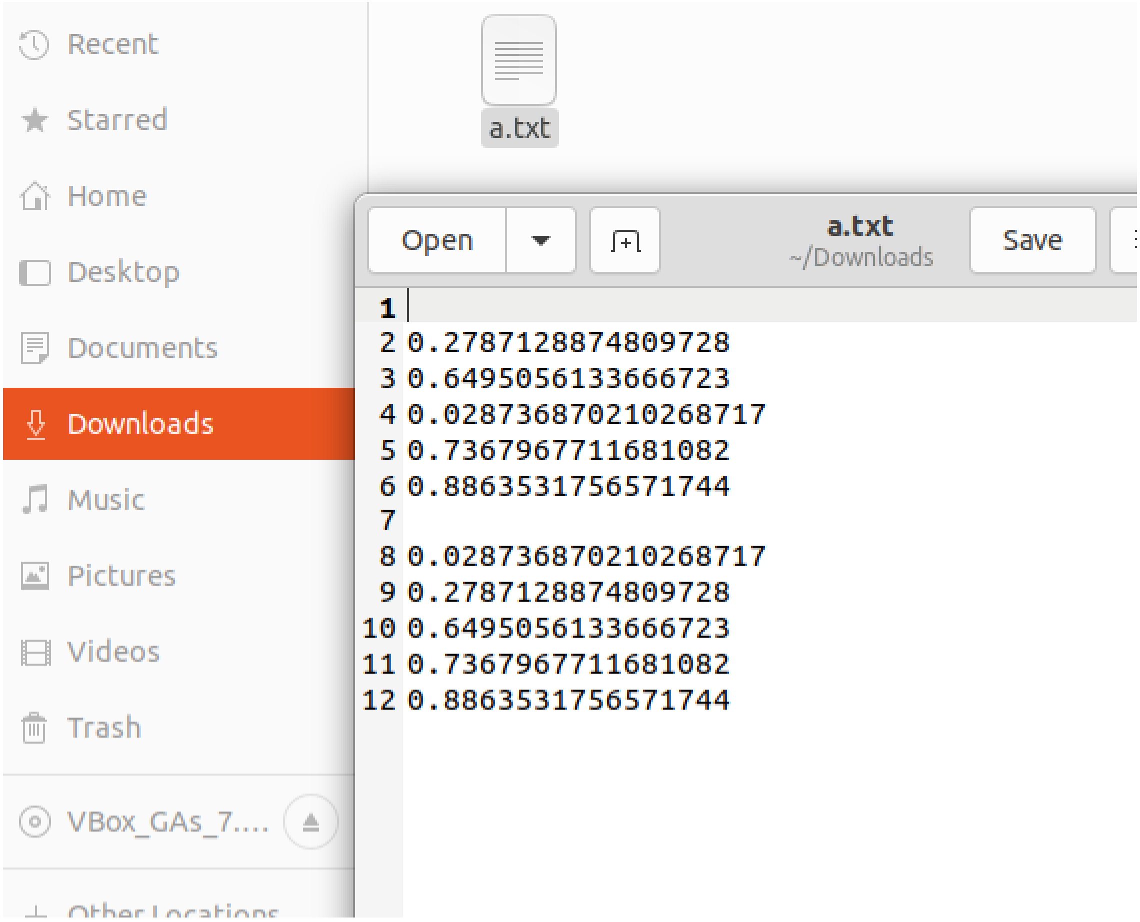
> `ifconfig` [to get new IP address]

> On FileZilla, enter the IP as host, username, password and port 21.



> transfer the file on FileZilla





c) In Linux, using terminal, use ls command to look for files from windows machine that has been uploaded **as question 3b above.**

```
> cd Downloads
> ls
```

FTP Security: Using chroot jail

4. A chroot on Unix Operating Systems is an operation that changes the apparent root directory for the current running process and its children. The programs that run in this modified environment cannot access the files outside the designated directory tree. This essentially limits their access to a directory tree and thus they get the name “chroot jail”.

a) Using ftp terminal command, identify your FTP server whether it implements chroot jail.

i) Login to FTP server and use pwd and ls command.

`> pwd`

```
ftp> pwd
257 "/home/osboxes" is the current directory
```

`> ls`

```
ftp> ls
200 PORT command successful
150 Opening ASCII mode data connection for file list
drwxrwxr-x   2 osboxes  osboxes      4096 Nov 27 14:36 car
drwxrwxr-x   2 osboxes  osboxes      4096 Nov 28 03:03 car1
drwxr-xr-x   2 osboxes  osboxes      4096 Nov 20 14:33 Desktop
drwxr-xr-x   2 osboxes  osboxes      4096 Nov 20 14:33 Documents
drwxr-xr-x   2 osboxes  osboxes      4096 Dec 12 02:59 Downloads
drwxr-xr-x   2 osboxes  osboxes      4096 Nov 20 14:33 Music
drwxr-xr-x   2 osboxes  osboxes      4096 Nov 20 14:33 Pictures
drwxr-xr-x   2 osboxes  osboxes      4096 Nov 20 14:33 Public
drwxr-xr-x   2 osboxes  osboxes      4096 Nov 20 14:33 Templates
drwxr-xr-x   2 osboxes  osboxes      4096 Nov 20 14:33 Videos
226 Transfer complete
```

ii) Go to root directory and use pwd and ls command.

`> cd /`

```
ftp> cd /
250 CWD command successful
```

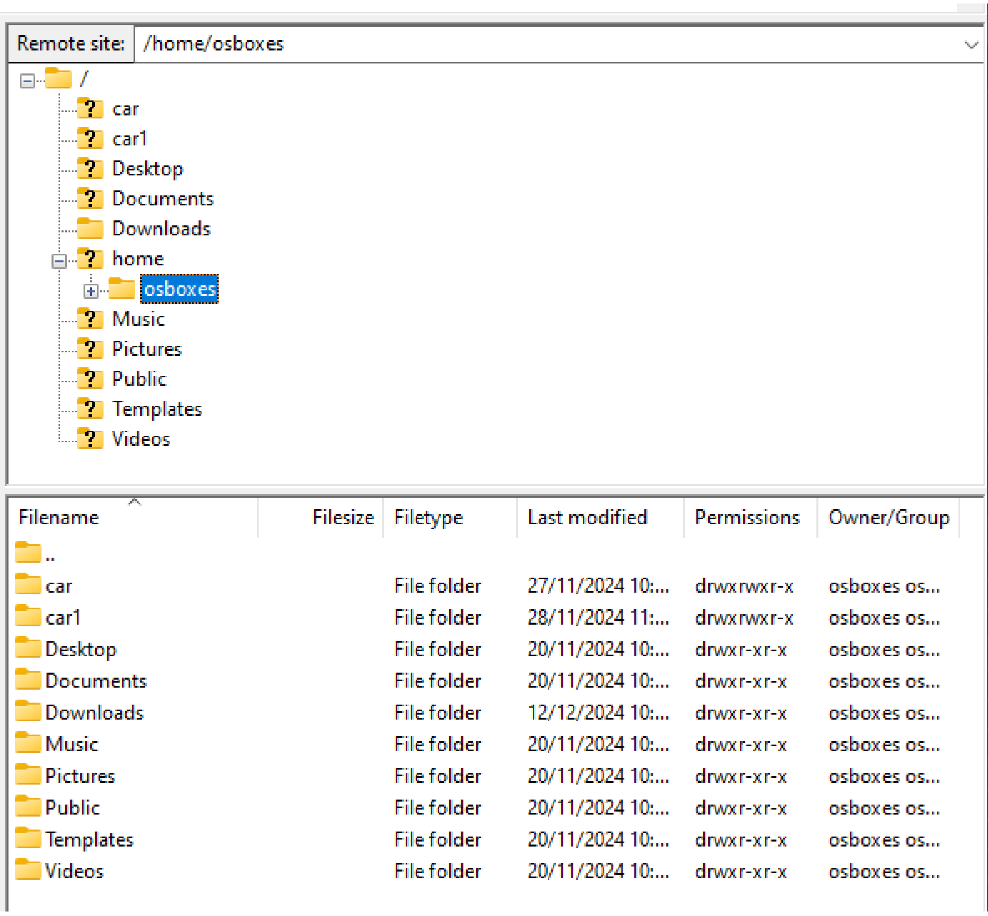
`> pwd`

```
ftp> pwd
257 "/" is the current directory
```

`> ls`

```
ftp> ls
200 PORT command successful
150 Opening ASCII mode data connection for file list
lrwxrwxrwx   1 root     root            7 Feb 23  2022 bin -> usr/bin
drwxr-xr-x   4 root     root         4096 Dec 12 02:18 boot
drwxr-xr-x  18 root     root         4080 Dec 12 02:49 dev
drwxr-xr-x 143 root     root        12288 Dec 11 13:48 etc
drwxr-xr-x   9 root     root         4096 Nov 28 02:53 home
lrwxrwxrwx   1 root     root            7 Feb 23  2022 lib -> usr/lib
lrwxrwxrwx   1 root     root            9 Feb 23  2022 lib32 -> usr/lib32
lrwxrwxrwx   1 root     root            9 Feb 23  2022 lib64 -> usr/lib64
lrwxrwxrwx   1 root     root           10 Feb 23  2022 libx32 -> usr/libx32
drwx------   2 root     root        16384 Mar 26  2022 lost+found
drwxr-xr-x   3 root     root         4096 Nov 20 14:44 media
drwxr-xr-x   2 root     root         4096 Feb 23  2022 mnt
drwxr-xr-x   3 root     root         4096 Nov 20 14:46 opt
dr-xr-xr-x 258 root     root            0 Dec 12 03:28 proc
drwx------   7 root     root         4096 Dec 12 02:49 root
drwxr-xr-x  37 root     root         1100 Dec 12 03:18 run
lrwxrwxrwx   1 root     root            8 Feb 23  2022 sbin -> usr/sbin
drwxr-xr-x   8 root     root         4096 Nov 21 02:56 snap
drwxr-xr-x   3 root     root         4096 Dec 11 13:47 srv
-rw-------   1 root     root      4121952256 Mar 26  2022 swap.img
dr-xr-xr-x  13 root     root            0 Dec 12 02:49 sys
drwxrwxrwt  17 root     root         4096 Dec 12 03:01 tmp
drwxr-xr-x  14 root     root         4096 Feb 23  2022 usr
drwxr-xr-x  14 root     root         4096 Nov 20 14:28 var
2 Show Applications plete
```

iii) Login to FTP using GUI client such as FileZilla.



iv) Give your comment on chroot jail implementation.

b) Edit the proftpd configuration sudo nano /etc/proftpd/proftpd.conf. Set the chroot jail by uncommenting the #DefaulRoot ~ and save/exit using CTRL+O and CTRL+X. Restart the FTP server using sudo systemctl restart proftpd.service.

> `sudo nano /etc/proftpd/proftpd.conf`

> Find line that writes "`#DefaultRoot ~`"

> Uncomment the line by removing the "`#`"

> Press `Ctrl + O`, then Enter to save

> Press `Ctrl + X` to exit

> Restart the ProFTPD service to apply changes

> `sudo systemctl restart proftpd.service`

```
osboxes@osboxes:~$ sudo nano /etc/proftpd/proftpd.conf


Use "fg" to return to nano.

[6]+  Stopped                 sudo nano /etc/proftpd/proftpd.conf
```

```
# Use this to jail all users in their homes
 DefaultRoot                     ~
```

```
osboxes@osboxes:~$ sudo systemctl restart proftpd.service
```

i) Login to FTP server and use pwd and ls command.

> `pwd`

> `ls`

```
ftp> pwd
257 "/" is the current directory
```

```
ftp> ls
200 PORT command successful
150 Opening ASCII mode data connection for file list
drwxrwxr-x   2 osboxes  osboxes      4096 Nov 27 14:36 car
drwxrwxr-x   2 osboxes  osboxes      4096 Nov 28 03:03 car1
drwxr-xr-x   2 osboxes  osboxes      4096 Nov 20 14:33 Desktop
drwxr-xr-x   2 osboxes  osboxes      4096 Nov 20 14:33 Documents
drwxr-xr-x   2 osboxes  osboxes      4096 Nov 20 14:33 Downloads
drwxr-xr-x   2 osboxes  osboxes      4096 Nov 20 14:33 Music
drwxr-xr-x   2 osboxes  osboxes      4096 Nov 20 14:33 Pictures
drwxr-xr-x   2 osboxes  osboxes      4096 Nov 20 14:33 Public
drwxr-xr-x   2 osboxes  osboxes      4096 Nov 20 14:33 Templates
drwxr-xr-x   2 osboxes  osboxes      4096 Nov 20 14:33 Videos
226 Transfer complete
```

ii) Go to root directory and use pwd and ls command.

> `cd /`

> `pwd`

```
ftp> pwd
257 "/" is the current directory
```

> ls

```
ftp> ls
200 PORT command successful
150 Opening ASCII mode data connection for file list
drwxrwxr-x    2 osboxes   osboxes       4096 Nov 27 14:36 car
drwxrwxr-x    2 osboxes   osboxes       4096 Nov 28 03:03 car1
drwxr-xr-x    2 osboxes   osboxes       4096 Nov 20 14:33 Desktop
drwxr-xr-x    2 osboxes   osboxes       4096 Nov 20 14:33 Documents
drwxr-xr-x    2 osboxes   osboxes       4096 Dec 12 02:59 Downloads
drwxr-xr-x    2 osboxes   osboxes       4096 Nov 20 14:33 Music
drwxr-xr-x    2 osboxes   osboxes       4096 Nov 20 14:33 Pictures
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drwxr-xr-x    2 osboxes   osboxes       4096 Nov 20 14:33 Templates
drwxr-xr-x    2 osboxes   osboxes       4096 Nov 20 14:33 Videos
226 Transfer complete
```
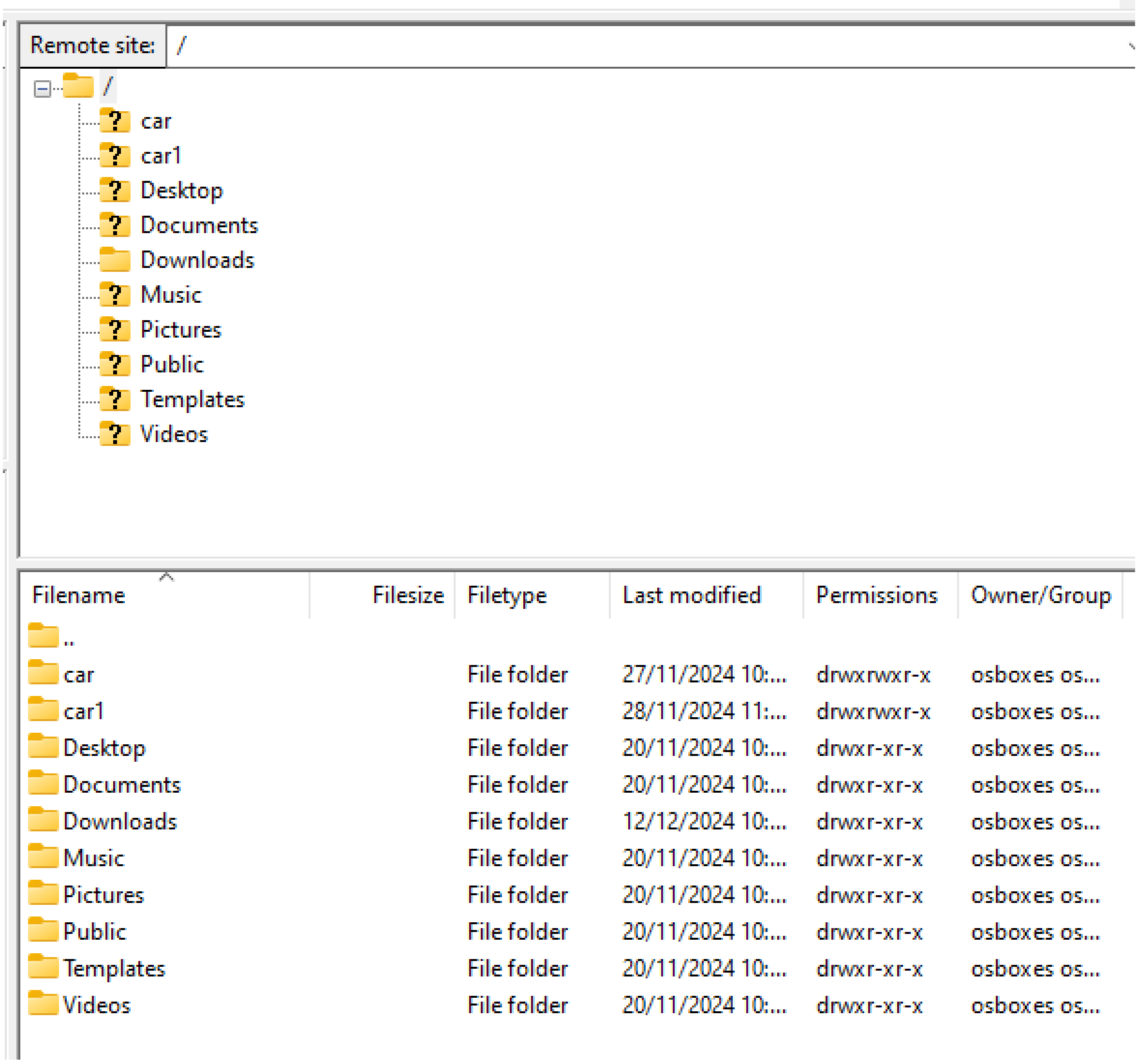

iii) Login to FTP using GUI client such as FileZilla.



iv) Give your comments.

Users are restricted to their home directories or specified paths. IF user attempts to navigate to /, it will not expose the actual root of the file system. The pwd command within the chroot jail will return paths relative to the jailed directory. This is to ensure that enhanced security, as users are not able to access sensitive system files or other users' data.